# STAT184 Final Project

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## 1 Introduction

This project explores trends in public confidence toward medicine using data from the General Social Survey (GSS). The analysis focuses on how trust in medical institutions changed between 2018 and 2024, particularly before and after the COVID-19 pandemic. Public trust in medicine is an important social indicator because it affects healthcare behavior, vaccine uptake, and confidence in scientific institutions.

## 2 Data Provenance

The dataset used in this project comes from the General Social Survey (GSS), a nationally representative survey conducted in the United States. The GSS collects demographic, behavioral, and attitudinal data from respondents across the country.

The main variables used in this analysis include:

- `CONMEDIC` — confidence in medicine
- `HAPPY` — self-reported happiness
- `PRAY` — frequency of prayer
- `CONCLERG` — confidence in clergy
- `CONFED` — confidence in the federal government

These variables allow the project to compare public confidence in medicine with other social attitudes and personal well-being measures.

# 3 FAIR and CARE Principles

## 3.1 FAIR Principles

The dataset partially satisfies the FAIR principles. It is **findable** because the GSS is publicly known and documented. It is **accessible** because researchers and students can access the data online. It is **interoperable** because it can be used in statistical software such as R. It is **reusable** because the GSS provides documentation and variable descriptions that support future analysis.

## 3.2 CARE Principles

The dataset only partially satisfies the CARE principles. The GSS provides broad demographic representation, but it was not specifically designed around Indigenous data governance, community ownership, or collective benefit. Because of this, the CARE principles are useful for thinking critically about the limits of the dataset.

# 4 Data Cleaning and Wrangling

The data required cleaning because missing or invalid values in the GSS are often represented with negative numbers. These values were converted into `NA` values before analysis. The analysis focuses on the years 2018, 2021, 2022, and 2024.

# 5 Exploratory Data Analysis

The exploratory analysis focuses on trends in medical confidence, happiness, and confidence in social institutions across multiple years.

## 5.1 Figure 1.1: Medical Confidence Table

Table 1: Medical confidence levels by year from the General Social Survey.

| Year | Medical Confidence | Count | Percent |
|---|---|---|---|
| 2018 | High Trust / Full Confidence | 553 | 35.6 |
| 2018 | Low / No Confidence | 211 | 13.6 |
| 2018 | Moderate / Skeptical | 790 | 50.8 |
| 2021 | High Trust / Full Confidence | 1070 | 40.2 |

| Year | Medical Confidence | Count | Percent |
|---|---|---|---|
| 2021 | Low / No Confidence | 246 | 9.2 |
| 2021 | Moderate / Skeptical | 1346 | 50.6 |
| 2022 | High Trust / Full Confidence | 786 | 33.4 |
| 2022 | Low / No Confidence | 373 | 15.8 |
| 2022 | Moderate / Skeptical | 1196 | 50.8 |
| 2024 | High Trust / Full Confidence | 569 | 26.4 |
| 2024 | Low / No Confidence | 428 | 19.8 |
| 2024 | Moderate / Skeptical | 1162 | 53.8 |

This table shows medical confidence categories by year from 2018 to 2024, including counts and percentages for high trust, moderate skepticism, and low confidence in medicine.

## 5.2 Figure 1.2: Stacked Bar Chart of Medical Confidence

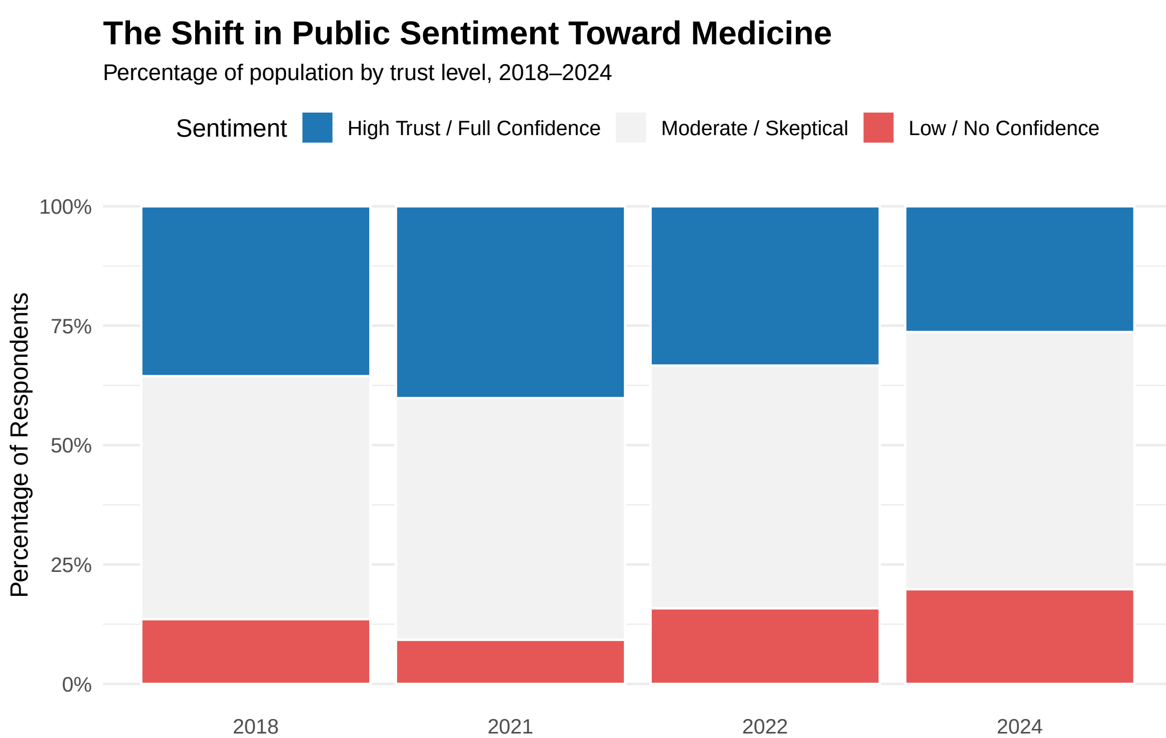



Figure 1: Stacked bar chart showing changes in public confidence toward medicine from 2018 to 2024.

The stacked bar chart shows the percentage distribution of trust levels in medicine for each year. This visualization makes it easier to compare how the balance of high trust, moderate skepticism, and low confidence changed over time.

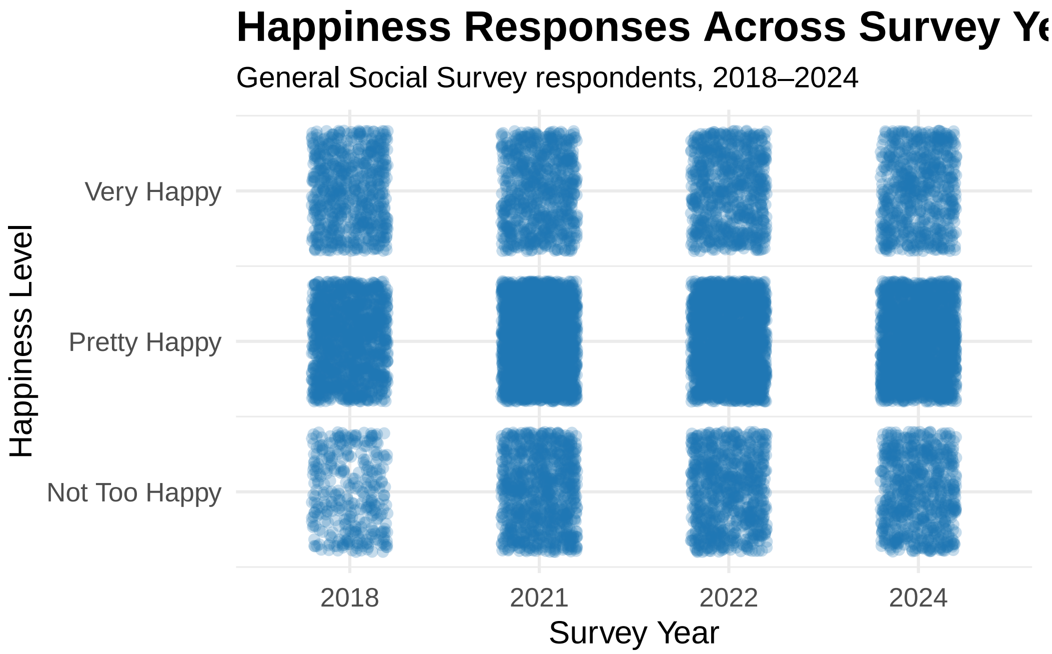
### 5.3 Figure 2: Mean Responses Table

Table 2: Mean Responses by Variable and Year

| Variable | 2018 | 2021 | 2022 | 2024 |
|---|---|---|---|---|
| Prayer | 1.43 | 1.63 | 1.58 | 1.53 |
| Happiness | 1.84 | 2.03 | 2.01 | 2.01 |
| Clergy Confidence | 2.05 | 2.18 | 2.21 | 2.21 |
| Govt Confidence | 2.33 | 2.32 | 2.38 | 2.34 |
| Health Confidence | 1.78 | 1.69 | 1.82 | 1.93 |

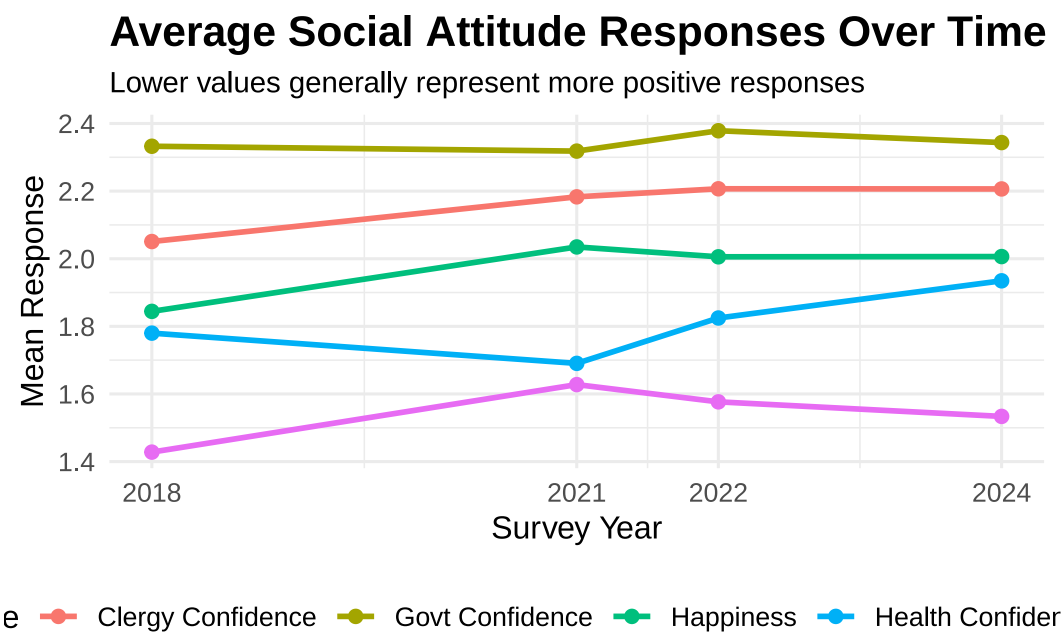
This table compares average responses across prayer frequency, happiness, confidence in clergy, confidence in government, and confidence in medicine. Lower values generally represent more positive responses for these variables. Prayer was divided by two so that it better matches the 1–3 scale of the other variables.

### 5.4 Figure 3: Happiness Scatterplot




The happiness scatterplot shows the spread of respondent happiness levels across the selected years. The jittered points help reduce overlap because happiness is measured using a small number of response categories.

### 5.5 Figure 4: Average Confidence and Happiness Trends




This line chart provides another view of how the selected variables changed over time. It helps compare medical confidence with happiness, prayer frequency, confidence in clergy, and confidence in the federal government.

## 6 Interpretation of Results

The visualizations suggest that public confidence in medicine changed across the selected survey years. The stacked bar chart gives a clearer view of the percentage of respondents in each medical confidence category, while the summary table and line chart show how medical confidence compares with other social attitudes. The happiness scatterplot adds context by showing how self-reported happiness varied during the same time period.

Overall, the analysis shows that confidence in institutions can be studied alongside measures of personal well-being and social behavior. These patterns are important because public trust in medicine and government can shape how people respond to major public health events and social changes.

# 7 Conclusion

This analysis used General Social Survey data to examine public confidence in medicine and related social variables from 2018 to 2024. Through data cleaning, exploratory analysis, professional tables, and multiple visualizations, the report highlights how public sentiment toward medical institutions and other social measures can be studied using R and Quarto.

# 8 Author Contributions

Abhiraj Srivastava was responsible for data cleaning, exploratory analysis, Quarto document organization, table creation, and visualization development.

Jonathan Hartman was responsible for the README.md, the GSSRepoPlan.docx, reviewing multiple code chunks, and creation of the happiness table and visualization.

Alexandre Snyder was responsible for the creation of the mean responses scatterplot, the happiness scatterplot, and reviewing all aforementioned work.

# 9 Code Appendix

This appendix contains the primary R code used for data cleaning, wrangling, table generation, and visualization throughout the analysis. Code is included to support reproducibility and transparency of the project workflow.

## 9.1 Package Loading and Data Import

```r
library(tidyverse)
library(ggplot2)
library(knitr)
library(here)

GSS_clean <- readRDS(here("GSS_clean.rds"))
```

## 9.2 Data Cleaning and Wrangling

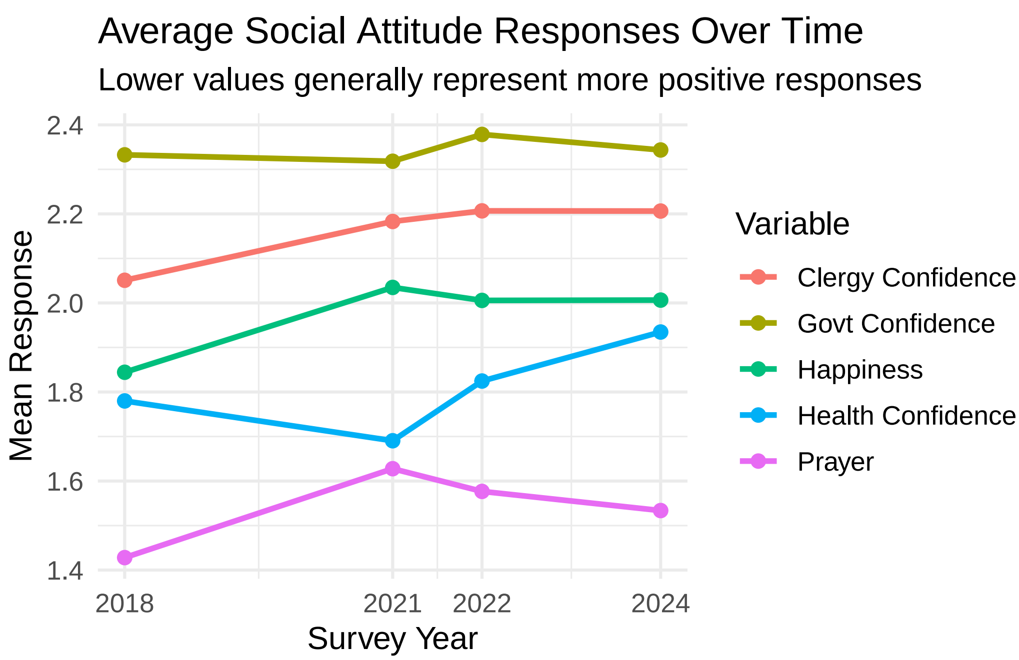
```r
exclude_vals <- c(
  -100, -99, -98, -97, -96,
  -95, -94, -93, -90, -80,
  -70, -60, -40
)

clean_data <- GSS_clean |>
  filter(YEAR %in% c(2018, 2021, 2022, 2024)) |>
  mutate(
    across(
      c(PRAY, HAPPY, CONCLERG, CONFED, CONMEDIC),
      ~ if_else(.x %in% exclude_vals, NA_real_, as.numeric(.x))
    )
  )

medical_data <- clean_data
```

## 9.3 Medical Confidence Table Code

```r
medical_table <- medical_data |>
  mutate(
    Medical_Confidence = case_when(
      CONMEDIC == 1 ~ "High Trust / Full Confidence",
      CONMEDIC == 2 ~ "Moderate / Skeptical",
      CONMEDIC == 3 ~ "Low / No Confidence",
      TRUE ~ NA_character_
    )
  ) |>
  drop_na(Medical_Confidence) |>
  count(YEAR, Medical_Confidence) |>
  group_by(YEAR) |>
  mutate(
    Percent = round(n / sum(n) * 100, 1)
  ) |>
  ungroup()
```

## 9.4 Visualization Code Example

```r
ggplot(
  trend_data,
  aes(
    x = YEAR,
    y = Mean,
    group = Variable,
    color = Variable
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  scale_x_continuous(
    breaks = c(2018, 2021, 2022, 2024)
  ) +
  labs(
    title = "Average Social Attitude Responses Over Time",
    subtitle = "Lower values generally represent more positive responses",
    x = "Survey Year",
    y = "Mean Response",
    color = "Variable"
  ) +
  theme_minimal(base_size = 12)
```




### 9.5 Session Information

```r
sessionInfo()
```

```
R version 4.5.3 (2026-03-11 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: America/New_York
tzcode source: internal
```

```
attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] here_1.0.2      knitr_1.51      lubridate_1.9.4 forcats_1.0.1
 [5] stringr_1.6.0   dplyr_1.2.0     purrr_1.2.1     readr_2.1.6
 [9] tidyr_1.3.2     tibble_3.3.1    ggplot2_4.0.2   tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] gtable_0.3.6       jsonlite_2.0.0     compiler_4.5.3     tidyselect_1.2.1
 [5] scales_1.4.0       yaml_2.3.12        fastmap_1.2.0      R6_2.6.1
 [9] labeling_0.4.3     generics_0.1.4     rprojroot_2.1.1    pillar_1.11.1
[13] RColorBrewer_1.1-3 tzdb_0.5.0         rlang_1.1.7        stringi_1.8.7
[17] xfun_0.57          S7_0.2.1           otel_0.2.0         timechange_0.3.0
[21] cli_3.6.5          withr_3.0.2        magrittr_2.0.4     digest_0.6.39
[25] grid_4.5.3         rstudioapi_0.18.0  hms_1.1.4          lifecycle_1.0.5
[29] vctrs_0.7.2        evaluate_1.0.5     glue_1.8.0         farver_2.1.2
[33] rmarkdown_2.31     tools_4.5.3        pkgconfig_2.0.3    htmltools_0.5.9
```